

Pakistani English Phonetic Inventory

This document presents the phonetic inventory used to represent Pakistani English. The inventory consists of two main categories of speech sounds. The first category includes sounds that occur in both Pakistani English and Standard English and are pronounced in a similar manner in the two varieties. The second category includes sounds that reflect the influence of Urdu, Pakistan's national language, on the pronunciation of Pakistani English. These Urdu-influenced sounds are incorporated into the inventory to represent pronunciation patterns that are characteristic of Pakistani English and may differ from those of Standard English.

1. Borrowed From Standard US & UK English

1.1 Consonants

Category	IPA Symbols
Stops	p, b, k, g
Nasals	m, n, ŋ
Fricatives	f, v, s, z, ʃ, ʒ, h
Affricates	tʃ, dʒ
Approximants	j, l, ɹ
Trill	r
Flap	ɾ

Note: Seven phones present in standard English are absent from the Pakistani English base inventory. The alveolar stops /t/ and /d/ are realized as ɽ and ɽ̌ respectively (retroflex; see Section 3). The dental fricatives θ and ð are realized as t̪ʰ and d̪ respectively. The labial-velar approximant /w/ is realized as /v/ (labiodental fricative). The diphthongs eɪ and oʊ are realized as the long monophthongs eː and oː respectively. All seven follow the phonemic collapse pattern inherited from Urdu phonology. See Section 3 for details.

1.2 Vowels and Diphthongs

Category	IPA Symbols
Short monophthongs	ɪ, ʊ, ə, ʌ, ɛ, æ, ɔ, ɒ
Long monophthongs	ɑː, iː, uː, ɜː, ɔː
Reduced / unstressed	ə
Diphthongs	aɪ (aɪ), aʊ (aʊ), ɔɪ

2. Pakistani English Additional Phones

2.1 Dental Consonants

Phone	Description	Example Word
t̪	Voiceless dental stop	tahira, pakistan
d̪	Voiced dental stop	sajid, saudi
t̪ʰ	Aspirated voiceless dental stop	tharparkar
d̪ʰ	Breathy-voiced dental stop	Dharampura, sindh

2.2 Retroflex Consonants

Phone	Description	Example Word
ɭ	Retroflex flap	Mohenjo-daro, Harappa, Jaranwala
ɖ	Voiced retroflex stop	David, added, damaged
ɖ	Voiceless retroflex stop	Khattak, Quetta, traffic, transport
ɖʰ	Aspirated voiceless retroflex stop	Thatta, Thakot
ɖʱ	Breathy-voiced retroflex stop	dhhol
ɭʱ	Breathy-voiced retroflex flap	karhi

2.3 Velar Fricatives

Phone	Description	Example Word
x	Voiceless velar fricative	khan , khushab
ɣ	Voiced velar fricative	ghalib , roghani

2.4 Common Aspirated Stops and Affricates

Phone	Description	Example Word
p ^h	Aspirated bilabial stop	phalia , phulkari
b ^h	Breathy-voiced bilabial stop	bhangra , bhera
k ^h	Aspirated velar stop	kharian
g ^h	Breathy-voiced velar stop	ghotki
tʃ ^h	Aspirated palato-alveolar affricate	chhattar
dʒ ^h	Breathy-voiced palato-alveolar affricate	jheenga , jhang

2.5 Uncommon Aspirated Sonorants

Phone	Description
m ^h	Breathy-voiced bilabial nasal
n ^h	Breathy-voiced alveolar nasal
l ^h	Breathy-voiced lateral approximant
r ^h	Breathy-voiced alveolar trill
j ^h	Breathy-voiced palatal approximant

2.6 Nasalized Vowels

Phone	Description
ã:	Nasalized long open back vowel
ẽ:	Nasalized long mid front vowel
ĩ:	Nasalized long close front vowel

Phone	Description
õ:	Nasalized long mid back vowel
ũ:	Nasalized long close back vowel
õ:	Nasalized long open-mid back vowel
æ̃:	Nasalized front raised vowel
ĩ	Nasalized short close front vowel
õ	Nasalized short close back vowel
ã	Nasalized schwa

2.7 Labiodental Approximant

Phone	Description
ʋ	Labiodental approximant

This phone is distinct from both the labiodental fricative /v/ and the labial-velar approximant /w/. It involves a lighter labiodental constriction than /v/, without the audible friction of a fricative, and is a genuine phone in Urdu-origin words rather than a target of the /w/→/v/ collapse described in Section 3.

2.8 Long Monophthongs

Phone	Description	Example Word
e:	Long mid front monophthong	gate, name
o:	Long mid back monophthong	home, road

3. Phonemic Collapses in Pakistani English

Pakistani English phonology inherits a systematic phonemic collapse from Urdu that parallels the reduction of Arabic-origin distinctions in Urdu itself. In Urdu, several sets of Arabic-origin consonants that are phonemically distinct in Arabic are realized identically:

س (seen), ث (say), ص (suad) → all realized as /s/

ز (ze), ذ (zal), ظ (zoe), ض (zuad) → all realized as /z/

ق (qaf) and ک (kaf) → both realized as /k/ in everyday speech

3.1 Alveolar Stop Collapse

Standard English does not distinguish place of articulation within its single set of coronal stops /t/ and /d/. Urdu, by contrast, contrasts a dental series (ت, د) with a distinct retroflex series (ٹ, ڈ). Pakistani English inherits these: every instance of standard English /t/ and /d/ is realized as either the dental t̪/d̪ (Section 2.1) or, more commonly for words without an inherited Urdu dental target, the retroflex ɖ/d̪ʁ (Section 2.2). No plain alveolar /t/ or /d/ occurs in Pakistani English.

In Standard English, /t/ and /d/ are alveolar stops, articulated with the tongue at the alveolar ridge. Urdu, by contrast, distinguishes between dental stops /t̪/ and /d̪/ (ت, د), articulated at the upper teeth, and retroflex stops /ɖ/ and /d̪ʁ/ (ٹ, ڈ), articulated with the tongue tip curled slightly backward. In the Pakistani English inventory described here, English /t/ and /d/ are realized as either dental or retroflex stops, reflecting the influence of Urdu. Dental realizations occur where an inherited Urdu pronunciation provides a corresponding dental target (Section 2.1); otherwise, retroflex realizations are more common (Section 2.2). Thus, plain alveolar /t/ and /d/ are not included as distinct phones in this inventory.

Standard English Phone	Pakistani English Realization
t (voiceless alveolar stop)	t̪ (voiceless retroflex stop)
d (voiced alveolar stop)	d̪ʁ (voiced retroflex stop)

So traffic is realized as /t̪ɾæfɪk/, transport as /t̪ɾænsˈpɔːt̪/, added as /æd̪ɪd̪/, and damaged as /d̪æmɪd̪ʁd̪/ in Pakistani English.

3.2 Dental Fricative Collapse

The voiceless dental fricative θ (as in "three", "thief", "think") and the voiced dental fricative δ (as in "the", "this", "that") do not occur as independent phones in Pakistani English.

Pakistani English speakers, regardless of education level, realize these sounds as the dental stops already present in the Urdu-influenced phoneme set:

Standard English Phone	Pakistani English Realization
θ (voiceless dental fricative)	t^h (aspirated voiceless dental stop)
δ (voiced dental fricative)	d (voiced dental stop)

This collapse is systematic and applies to all Pakistani English speakers. A speaker who produces θ and δ is aiming toward native English pronunciation rather than Pakistani English.

3.3 Labial Approximant Collapse

A parallel collapse applies to the labial-velar approximant /w/. In Urdu, the letter و (waw) is realized as a labiodental sound, and Pakistani English speakers consistently produce the labiodental fricative /v/ for both /v/ and /w/ words of standard English. The rounded-lip /w/ articulation does not occur in Pakistani English. Pakistani English speakers produce labiodental contact (upper teeth on lower lip) for both. This means:

Standard English Phone	Pakistani English Realization
w (labial-velar approximant)	v (labiodental fricative)

So water is realized as /vætər/, we as /vi:/, and walk as /vɔ:k/ in Pakistani English.

3.4 Diphthong Monophthongization

A further collapse applies to the diphthongs eɪ and oʊ. These diphthongs are absent from the vowel systems of Urdu, Punjabi, Sindhi, and Pashto. Consequently, even highly educated Pakistani English speakers systematically realize them as long monophthongs, holding the first vowel position rather than gliding:

Standard English Phone	Pakistani English Realization
eɪ (diphthong)	e: (long mid front monophthong)
oʊ/əʊ (diphthong)	o: (long mid back monophthong)

So gate is realized as /ge:t/, name as /ne:m/, home as /ho:m/, and road as /ro:d/ in Pakistani English. Notably, espeak en-pk does not implement this monophthongization and retains eɪ and əʊ in its output.

The text analysis module applies all seven collapses as post-processing steps on espeak en-us output: $t \rightarrow t$, $d \rightarrow d$, $\theta \rightarrow t^h$, $\delta \rightarrow d$, $w \rightarrow v$, $eɪ \rightarrow e:$, and $oʊ/\əʊ \rightarrow o:$. All seven are inherited from Urdu phonology, and the collapsed source phones (t, d, θ, δ, w, eɪ, oʊ) are excluded from the Pakistani English phonetic inventory accordingly.

4. Complete Pakistani English Phoneme Set

The full Pakistani English phoneme inventory is the union of that borrowed from the standard English base inventory (Section 1) and the Pakistani English additional phones (Section 2), with the phonemic collapses described in Section 3 applied. This combined set defines the symbol space for both the TTS system and the text normalization pipeline.

Total phones: Standard English base (21 consonants + 17 vowels = 38 phones) + Pakistani English additional (36 phones) = 74 phones